

Public and Private - A public ledger allows anyone to read through it. Private, on the other hand, is a little exclusive in nature and only allows a select few to peek. Both follow the principles of distributed verification but differentiate in the number of mining nodes. Also, a public ledger due to security measurements is slower than private.

Permissionless and Permissioned - A permissionless system is an open system that does not maintain a set of requirements for becoming a node and therefore allows anyone to join in. Permissioned, on the other hand, maintains rules that decide who can become a node and who cannot. These permissions define the stakes of the nodes, therefore, reducing the risk of malicious behaviour.

Two-axis blockchain classification - *Public-Private* and *Permissionless-Permissioned* are used together to classify a blockchain. A permissionless public blockchain allows anyone to become a node and validate the transactions. In a permissioned public blockchain, anyone who meets certain requirements can become a node. Permissionless private systems only allow a specific type of people, but because there are no requirements, the stakes are reduced for each member. Finally, a permissioned private blockchain only allows a specific group of people, usually very small and all of them undergo a set of requirements to increase their stakes.

The Three-Headed Beast

As of the writing of this article, a blockchain can be one of three broad categories. Public, Private and Federated. Public blockchains are the purest form of the technology. Anyone can join the chain, read it, create a node, mine and validate transactions. There is no centralization whatsoever. Though some of the public blockchains have minor restrictions, they still welcome anyone and are distributed. Private blockchains, on the other hand, are on the other end of the spectrum. They are managed by either a small group of people. And they decide who can read through the blocks, create a node and operate on it. They still follow the encrypted nature of pure blockchains. Finally, the federated blockchains are the middle ground, sort of. Instead of containing all the management into one person or a group of people, a federation or a consortium of organisations together handle management. But the fact that the last two variants are centralized, defeats the purpose of blockchains. So in order to remove confusion another term, distributed ledger technology, was coined. This term umbrellas all the three forms of a blockchain.